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1. Introduction

Scalar transport on the sphere is widely used in atmospheric and geophysical applications, while longitude–latitude coordinates suffer from singularities at the poles. To avoid this issue, we adopt an octahedron-based triangular manifold mesh.

Since quadrature-nodal DG discretizations generally do not preserve the continuous chain rule or product rule, a split formulation is introduced to correct the discrete structure of the conservation form. Its numerical behavior is then evaluated through the time evolution of mass and L2 errors.

2. Mathematical Formulation

The mapping conservation law is written as

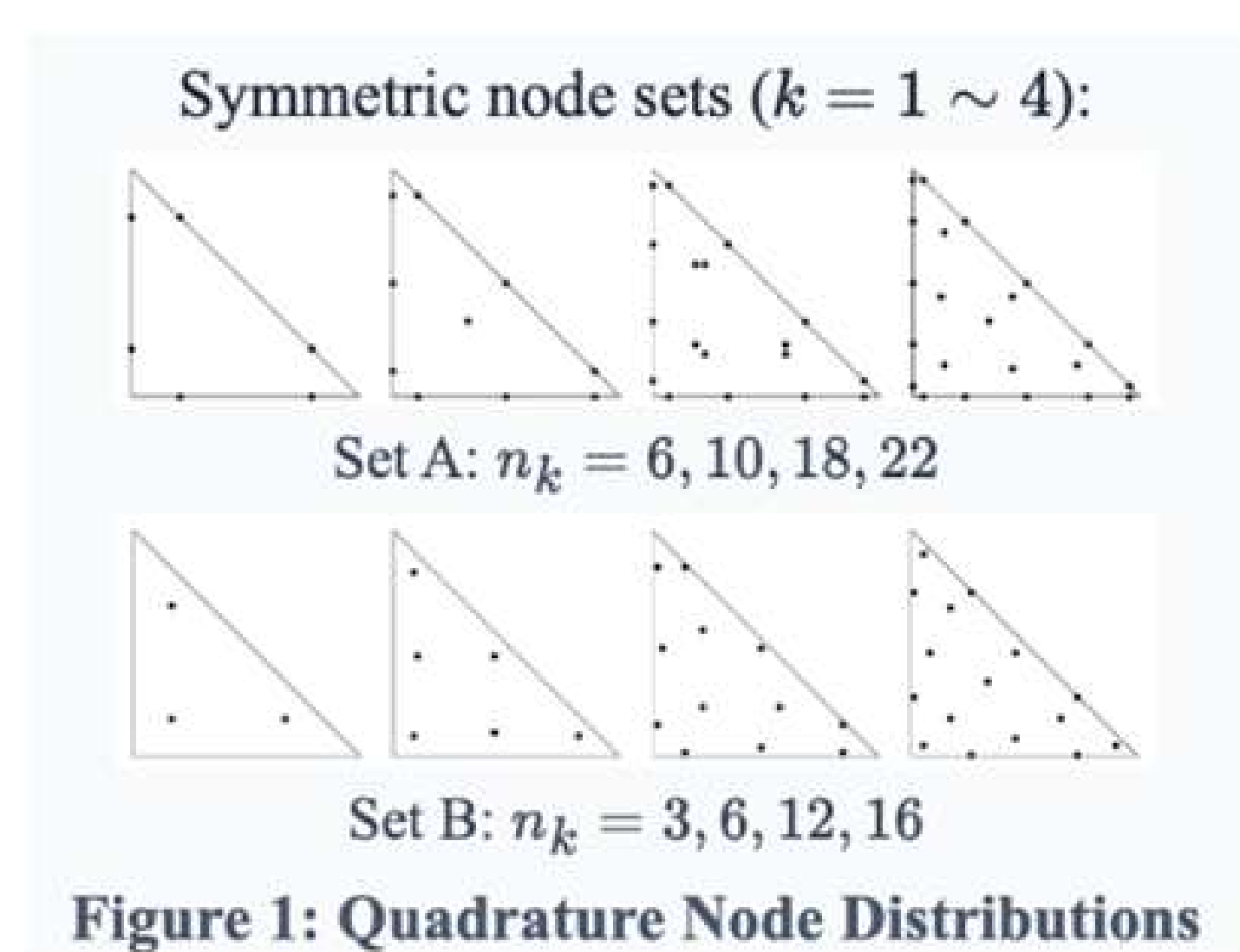
$$\frac{\partial(Jq)}{\partial t} + \frac{\partial(Ju^r q)}{\partial r} + \frac{\partial(Ju^s q)}{\partial s} = 0.$$

Here, J is the surface Jacobian, while u^r and u^s are the contravariant velocity components.

The mapped conservation law can be rewritten as the split form

$$J \frac{\partial q}{\partial t} + \frac{1}{2} \left[\frac{\partial(Ju^r q)}{\partial r} + Ju^r \frac{\partial q}{\partial r} + q \frac{\partial(Ju^r)}{\partial r} + \frac{\partial(Ju^s q)}{\partial s} + Ju^s \frac{\partial q}{\partial s} + q \frac{\partial(Ju^s)}{\partial s} \right] = 0.$$

3. Mesh Mapping & Field Setup



這是映射

這是初始場

這是風場

References

4. Discrete Formulation

Quadrature-Nodal Operators

$$V_{ij} = \phi_j(r_i, s_i), \quad (V_b^\gamma)_{ij} = \phi_j(r_i^\gamma, s_i^\gamma), \quad (V_m)_{ij} = \frac{\partial \phi_j(r_i^\gamma, s_i^\gamma)}{\partial m}, \quad P = (V^T W V)^{-1} V^T W,$$

$$D_r = V_r P, \quad D_s = V_s P, \quad R^\gamma = V_b^\gamma P, \quad L^\gamma = \frac{1}{J} W^{-1} (R^\gamma)^T W_b.$$

Split-Form DG Discretization

$$\frac{d\mathbf{q}}{dt} = -\frac{1}{2J} \left[D_r((Ju^r) \odot \mathbf{q}) + (Ju^r) \odot D_r \mathbf{q} + \mathbf{q} \odot D_r(Ju^r) + D_s((Ju^s) \odot \mathbf{q}) + (Ju^s) \odot D_s \mathbf{q} + \mathbf{q} \odot D_s(Ju^s) \right] + \sum_{\gamma \in \partial T} L^\gamma p^\gamma.$$

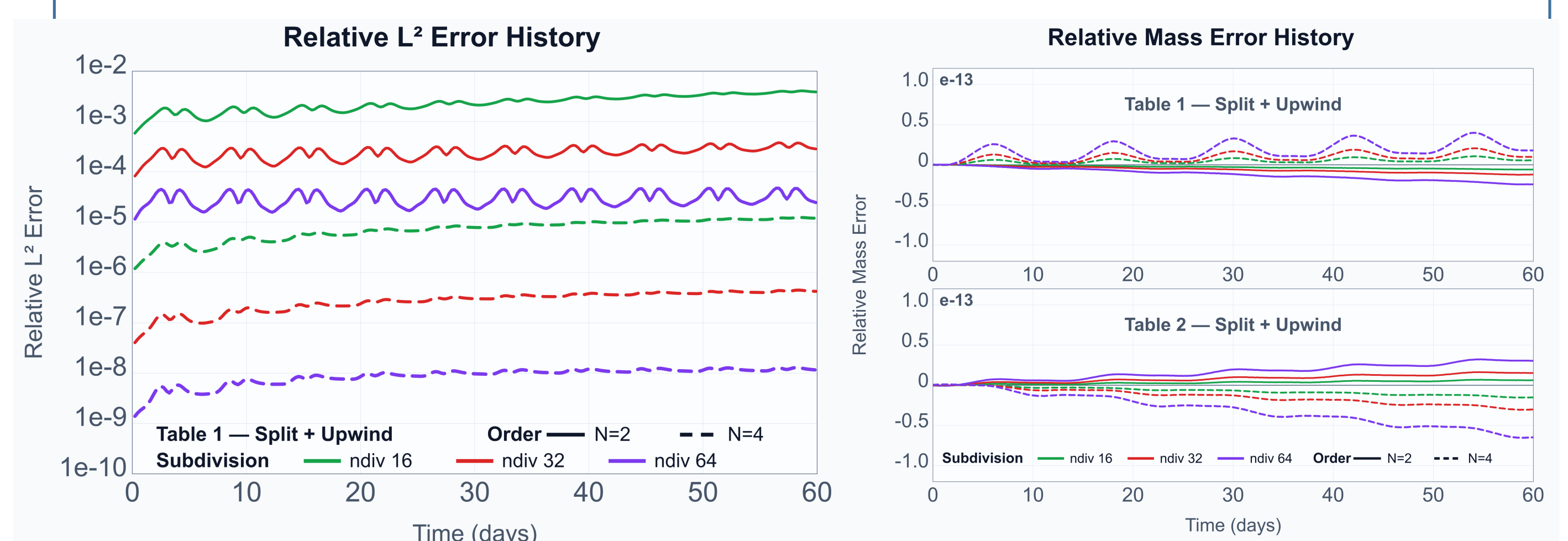
$$p^\gamma = \frac{1}{2} \left[n_r^\gamma R^\gamma (Ju^r \odot \mathbf{q}) + n_s^\gamma R^\gamma (Ju^s \odot \mathbf{q}) + [n_r^\gamma R^\gamma (Ju^r) + n_s^\gamma R^\gamma (Ju^s)] \odot (R^\gamma \mathbf{q}) \right] - F^{*,\gamma}.$$

Time Integration

$$\frac{d\mathbf{q}}{dt} = \mathcal{R}(\mathbf{q}, t), \quad \boxed{\text{LSRK54 : 5-stage, 4th-order}}$$

5. Numerical Results

誤差計算方式 收斂階計算方式



6. Conclusion